

CHE 201 Fall 2019 HW 3

Alan Gutierrez

TOTAL POINTS

78 / 100

QUESTION 1

11 15 / 15

- ✓ - **0 pts** Correct
- **7 pts** b. was missing
- **2 pts** $W = -2\text{KJ}$; $\Delta PE = 2\text{KJ}$
- **7 pts** b. incorrect

QUESTION 2

22 15 / 15

- ✓ - **0 pts** Correct
- **3 pts** unit conv for Rotation
- **5 pts** c. was incorrect.

QUESTION 3

33 20 / 20

- ✓ - **0 pts** Correct
- **5 pts** Wrong solution for work (calculation and/or units conversion error). Correct answer is $\text{Work} = 30.794 \text{ Btu/lb}$
- **5 pts** Correct equation, but wrong solution for T_2 . Correct answer is 819.9 oR
- **2.5 pts** $\Delta u = -W = -30.79 \text{ Btu/lb}$; Correct answer for T_2 is 819.9 oR
- **2.5 pts** Missing energy balance
- **20 pts** Wrong solution
- **5 pts** Did not state the postulates applied to solve the problem

QUESTION 4

44 15 / 15

- ✓ - **0 pts** Correct
- **5 pts** Wrong equation for coefficient of performance. The correct answer is 3
- **15 pts** Wrong answer. $Q_{in} = 0.45 \text{ kw}$ and $\beta = 3$

QUESTION 5

55 13 / 15

- **0 pts** Correct
- **15 pts** Wrong or missing answer
- **4 pts** $V_a = 9.94 \times 10^{-3} \text{ m}^3/\text{kg}$
- **4 pts** $V_b = 0.9992 \times 10^{-3} \text{ m}^3/\text{kg}$
- **4 pts** $V_c = \text{aprox } 1.0078 \times 10^{-3} \text{ m}^3/\text{kg}$ ($= V_f \text{ sat at } 40 \text{ oC}$)
- **3 pts** Wrong or missing sketch of T-v diagram
- **1 pts** b and c should be at the same temperature (sketch)
- ✓ - **2 pts** Should not extrapolate the table limits. $V_c = \text{aprox } 1.0078 \times 10^{-3} \text{ m}^3/\text{kg}$ ($= v_{fsat} \text{ at } 40\text{oC}$)
- **1 pts** Point c should be outside the L-V region (sketch)
- **1 pts** point b should be outside the L-V region (sketch), in the compressed liquid region
- **2 pts** $V_a = 9.94 \times 10^{-3} \text{ m}^3/\text{kg}$ (superheated vapor - Table A-4)
- **2 pts** V_b missing the term $\times 10^{-3}$
- **1 pts** Point a should be above the critical point
- **1 pts** c) P should be 2Mpa and not 20MPa in the sketch of the T-v diagram

QUESTION 6

66 0 / 20

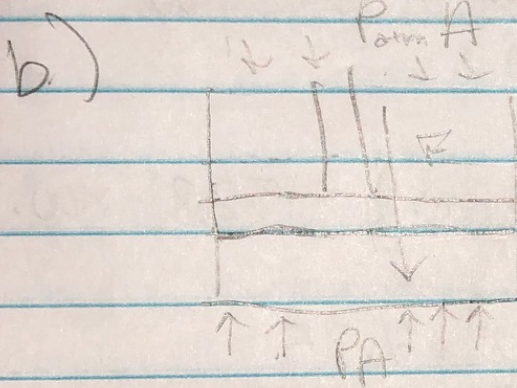
- **0 pts** Correct
- **5 pts** Did not properly describe the phase-out schedule (action and deadlines) for R-22 in the US
- **5 pts** Did not list the refrigerants being used to replace R-22
- **5 pts** Did not include at least 3 references
- **3 pts** Did not properly list the references used
- **2 pts** Did not properly list the refrigerants being used to replace R-22
- ✓ - **20 pts** Wrong or Missing answer
- **10 pts** Did not describe the phase-out schedule

(action and deadlines) for R-22 in the US

H.W. #3

1.) a.) $W = p(V_2 - V_1)$
 $W = (2 \text{ bar}) \left(\frac{10^5 \text{ Pa}}{1 \text{ bar}} \right) (0.12 \text{ m}^3 - 0.1 \text{ m}^3)$
 $= 4000 \text{ N}\cdot\text{m} \times \frac{1 \text{ kJ}}{10^3 \text{ N}\cdot\text{m}} = \boxed{4 \text{ kJ}}$

$Q = \Delta U + W$
 $= 0.25 \text{ kJ} + 4 \text{ kJ}$
 $= \boxed{4.25 \text{ kJ}}$



$F = (P_{\text{atm}} - p)A$
 $W = (P_{\text{atm}} - p)(V_2 - V_1)$

$W = (1-2) \text{ bar} \times \frac{100 \text{ kPa}}{1 \text{ bar}} \times (0.12 \text{ m}^3 - 0.1 \text{ m}^3)$
 $= -2 \text{ kN}\cdot\text{m}^3/\text{m}^2 \times \frac{1 \text{ kJ}}{1 \text{ kN}\cdot\text{m}} = \boxed{-2 \text{ kJ}}$

$Q - W = \Delta E$

$0 - (-2) = 0 + 0 + \Delta PE$

$\boxed{2 \text{ kJ} = \Delta PE}$

11 15 / 15

✓ - 0 pts Correct

- 7 pts b. was missing

- 2 pts $W = -2\text{KJ}$; $\Delta PE = 2\text{KJ}$

- 7 pts b. incorrect

2) $V = 110 \text{ V}$ $\omega = \frac{(2\pi)(2000)}{60}$
 $I = 20 \text{ amp}$
 $t = 9.07 \text{ Nm}$ $= 209.44 \text{ rad/rev}$
 $s = 2000 \text{ RPM}$

a) $P_{in} = VI = (110)(20) = \boxed{2200 \text{ W}}$

b) $P_{out} = t\omega = (9.07)(209.44) = \boxed{1899.62 \text{ W}}$

c) $2200 \text{ W} \times \frac{1 \text{ kW}}{10^3 \text{ W}} = 2.2 \text{ kW}$

$P_{in} = (2.2 \text{ kW})(24 \text{ hrs}) = 52.8 \text{ kW} \cdot \text{hrs}$

$(0.09)(52.8) = \boxed{\$4.75}$

2 2 15 / 15

✓ - 0 pts Correct

- 3 pts unit conv for Rotation

- 5 pts c. was incorrect.

#3

$$V_2 = V_1 \left(\frac{P_1}{P_2} \right)^{1/n}$$

$$P_1 = 100 \text{ lbf/in}^2$$

$$V_1 = 3.704 \text{ ft}^3/\text{lb}$$

$$T_1 = 1000^\circ \text{R}$$

$$P_2 = 50 \text{ lbf/in}^2$$

$$V_2 = ? = 6.077 \text{ ft}^3/\text{lb}$$

$$n = 1.4$$

$$T_2 = ?$$

$$W = \frac{P_2 V_2 - P_1 V_1}{1 - n}$$

$$= \frac{(50 \text{ lbf/in}^2)(6.077 \text{ ft}^3/\text{lb}) - (100 \text{ lbf/in}^2)(3.704 \text{ ft}^3/\text{lb})}{1 - 1.4}$$

$$\frac{166.375 \text{ ft}^3 \cdot \text{lbf}}{\text{lb} \cdot \text{in}^2} \times \frac{144 \text{ in}^2}{1 \text{ ft}} \times \frac{1 \text{ Btu}}{778 \text{ ft} \cdot \text{lbf}} = 30.794 \text{ Btu/lb}$$

$$0 = \Delta U + W$$

$$\Delta U = -W$$

$$C_v (T_2 - T_1) = -W$$

$$(0.171)(T_2 - 1000) = -(30.794)$$

$$0.171 T_2 - 171 = -30.794$$

$$0.171 T_2 = 140.206$$

$$T_2 = 819.92^\circ \text{R}$$

33 20 / 20

✓ - 0 pts Correct

- 5 pts Wrong solution for work (calculation and/or units conversion error). Correct answer is Work = 30.794

Btu/lb

- 5 pts Correct equation, but wrong solution for T2. Correct answer is 819.9 oR

- 2.5 pts $\Delta u = -W = -30.79$ Btu/lb; Correct answer for T2 is 819.9 oR

- 2.5 pts Missing energy balance

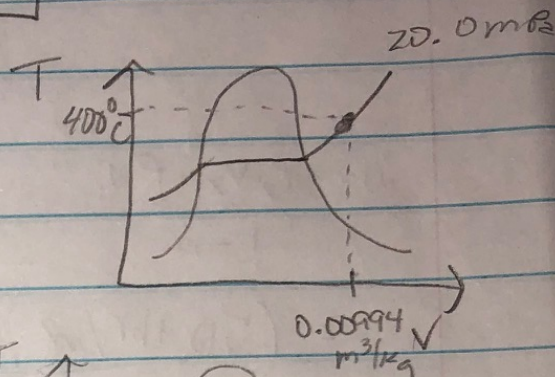
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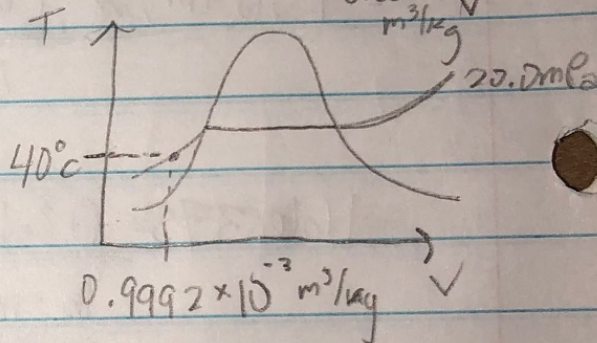
4. $\dot{W}_{\text{cycle}} = \dot{Q}_{\text{out}} - \dot{Q}_{\text{in}}$
 $\dot{Q}_{\text{in}} = (0.6 \text{ kW}) - (0.15 \text{ kW})$
 $= \boxed{0.45 \text{ kW}}$

$\beta = \frac{\dot{Q}_{\text{in}}}{\dot{W}_{\text{cycle}}} = \frac{0.45}{0.15} = \boxed{3}$

5. a) $p = 20 \text{ MPa}$
 $T = 400^\circ\text{C}$
 $v = 0.00994 \text{ m}^3/\text{kg}$



b) $p = 20 \text{ MPa}$
 $T = 40^\circ\text{C}$
 $v = 0.9992 \times 10^{-3} \text{ m}^3/\text{kg}$



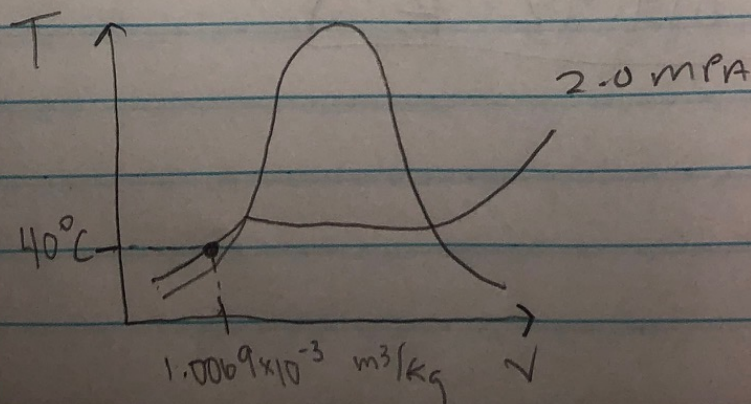
c) $v = v_{2.5 \text{ MPa}} + (v_{5 \text{ MPa}} - v_{2.5 \text{ MPa}}) \left(\frac{p - p_{2.5 \text{ MPa}}}{p_{5 \text{ MPa}} - p_{2.5 \text{ MPa}}} \right)$

$= 1.0067 \times 10^{-3} + (1.00056 \times 10^{-3} - 1.00067 \times 10^{-3}) \left(\frac{2 - 2.5}{5 - 2.5} \right)$

$v = 1.0069 \times 10^{-3} \text{ m}^3/\text{kg}$

$p = 2 \text{ MPa}$

$T = 40^\circ\text{C}$



4 4 15 / 15

✓ - 0 pts Correct

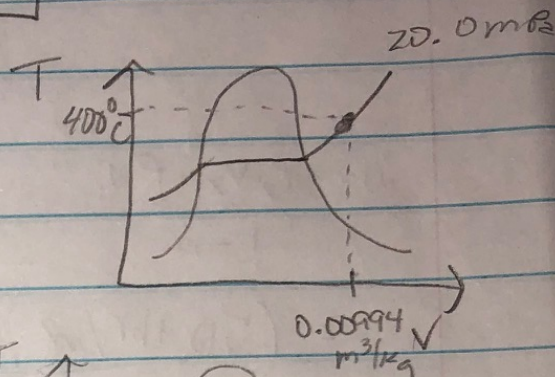
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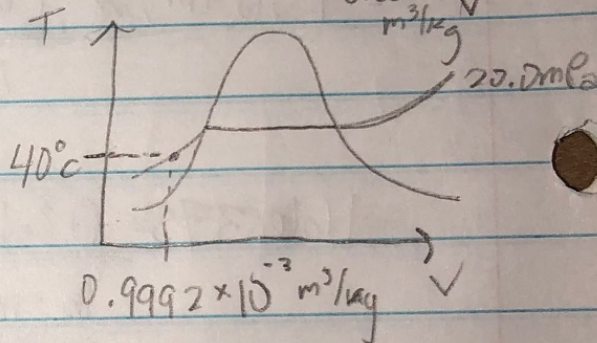
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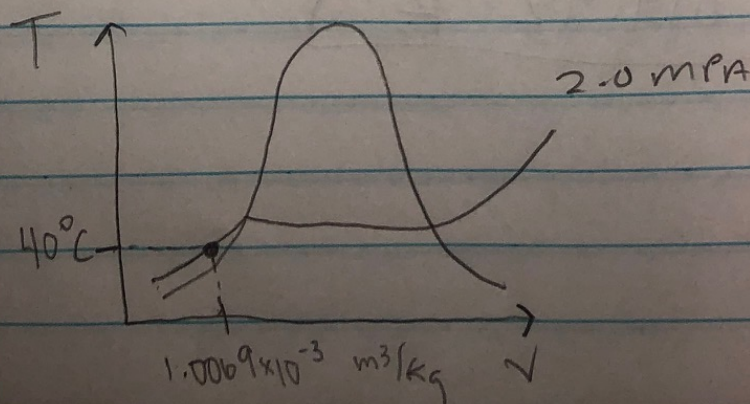
c) $v = v_{2.5 \text{ MPa}} + (v_{5 \text{ MPa}} - v_{2.5 \text{ MPa}}) \left(\frac{p - p_{2.5 \text{ MPa}}}{p_{5 \text{ MPa}} - p_{2.5 \text{ MPa}}} \right)$

$= 1.0067 \times 10^{-3} + (1.00056 \times 10^{-3} - 1.00067 \times 10^{-3}) \left(\frac{2 - 2.5}{5 - 2.5} \right)$

$v = 1.0069 \times 10^{-3} \text{ m}^3/\text{kg}$

$p = 2 \text{ MPa}$

$T = 40^\circ\text{C}$



5. $P_1 = P_{\text{atm}} + (P_{\text{gage}})_1$

$$P_1 = 401.325 \text{ kPa}$$

$$P_2 = P_{\text{atm}} + (P_{\text{gage}})_2$$

$$P_2 = 468.325$$

$$(P_{\text{gage}})_1 = 300 \text{ kPa}$$

$$P_{\text{atm}} = 101.325 \text{ kPa}$$

$$(P_{\text{gage}})_2 = 367 \text{ kPa}$$

$$T_1 = 300 \text{ K}$$

$$T_2 = \frac{P_2}{P_1} T_1$$

$$= \frac{468.325}{401.325} (300 \text{ K})$$

$$= 350.084 \text{ K} = 77.084 ^\circ\text{C}$$

5 5 13 / 15

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$$= 350.084 \text{ K} = 77.084 ^\circ\text{C}$$

660 / 20

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